


**DEPARTMENT OF INFORMATION TECHNOLOGY**

**24ITT32 - PROGRAMMING IN JAVA**

# INDIVIDUAL PROJECT ON
# ACADEMIC MANAGEMENT SYSTEM

**P R O J E C T  R E P O R T**

Submitted by

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# JAVAASSIGNMENT

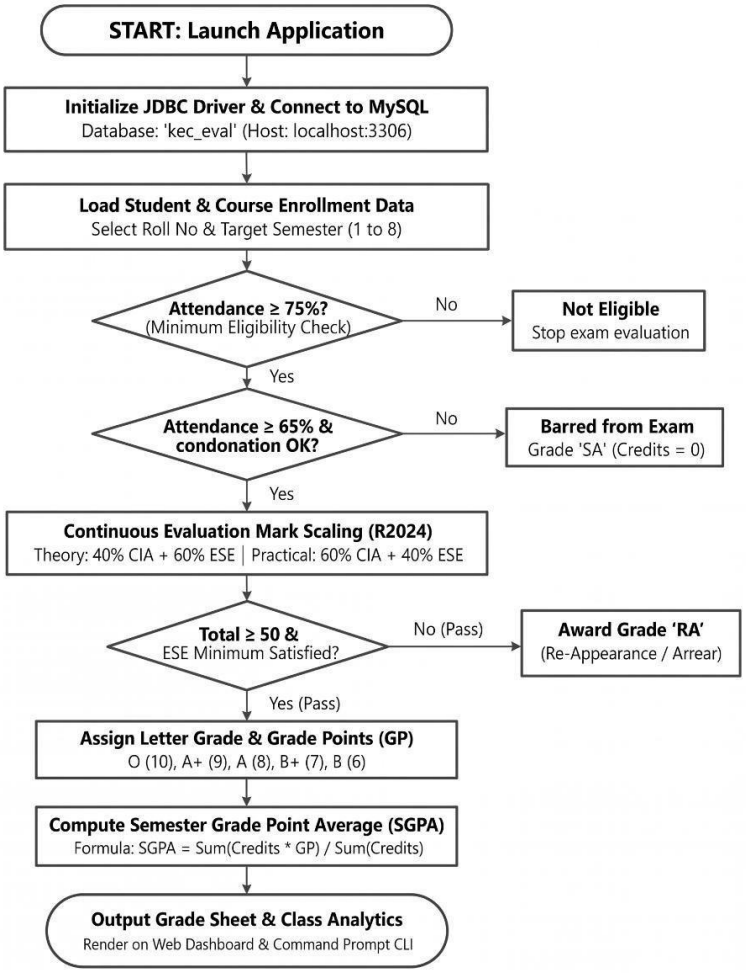
## AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

## OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

## FLOW CHART:



## PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

## DATABASE SCHEMA:

### 1. STUDENTS TABLE:

```
mysql> describe students;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| roll_no          | varchar(10)  | NO   | PRI | NULL    |       |
| student_name     | varchar(100) | NO   |     | NULL    |       |
| department       | varchar(10)  | NO   |     | NULL    |       |
| current_semester | int          | NO   |     | NULL    |       |
| email            | varchar(100) | YES  |     | NULL    |       |
| phone            | varchar(15)  | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

### 2. COURSES TABLE:

```
mysql> describe courses;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| course_code | varchar(10)  | NO   | PRI | NULL    |       |
| title       | varchar(100) | NO   |     | NULL    |       |
| credits     | decimal(3,1) | NO   |     | NULL    |       |
| course_type | varchar(10)  | NO   |     | NULL    |       |
| department  | varchar(10)  | NO   |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

### 3. ENROLLMENTS TABLE:

```
mysql> describe enrollments;
+------------------+-------------+------+-----+---------+----------------+
| Field            | Type        | Null | Key | Default | Extra          |
+------------------+-------------+------+-----+---------+----------------+
| enrollment_id    | int         | NO   | PRI | NULL    | auto_increment |
| roll_no          | varchar(10) | NO   | MUL | NULL    |                |
| course_code      | varchar(10) | NO   | MUL | NULL    |                |
| semester         | int         | NO   |     | NULL    |                |
| classes_held     | int         | NO   |     | NULL    |                |
| classes_attended | int         | NO   |     | NULL    |                |
+------------------+-------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

## 4. ASSESSMENT MARKS TABLE:

```
mysql> describe assessment_marks;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| mark_id         | int          | NO   | PRI | NULL    | auto_increment |
| enrollment_id   | int          | NO   | MUL | NULL    |                |
| cia1            | decimal(5,2) | YES  |     | NULL    |                |
| cia2            | decimal(5,2) | YES  |     | NULL    |                |
| assignment_mark | decimal(5,2) | YES  |     | NULL    |                |
| ese_mark        | decimal(5,2) | YES  |     | NULL    |                |
| ese_absent      | tinyint(1)   | YES  |     | 0       |                |
+-----------------+--------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

# SOURCE CODE:

## MAINAPP.JAVA:

```
import java.sql.Connection;

import java.sql.SQLException;

import java.util.Scanner;


public class MainApp {

    private static final Scanner sc = new Scanner(System.in);

    private static final StudentDAO students = new StudentDAO();

    private static final AcademicDAO academic = new AcademicDAO();


    public static void main(String[] args) {

        try (Connection c = DBConnection.getConnection()) {

            System.out.println("Database connection successful.");

        } catch (SQLException e) {

            System.out.println("Database connection failed: " + e.getMessage());

            System.out.println("Check MySQL is running and update DBConnection.java password.");

            return;

        }


        while (true) {

            System.out.println("\n=== KEC STUDENT ACADEMIC & CONTINUOUS EVALUATION SYSTEM

===");
```

```
            System.out.println("1. Add Student");
            System.out.println("2. Add Course");
            System.out.println("3. Enroll Student in Course");
            System.out.println("4. Update Attendance");
            System.out.println("5. Enter/Update Incremental Marks");
            System.out.println("6. Generate Semester Mark Sheet");
            System.out.println("7. Attendance Defaulter Report");
            System.out.println("0. Exit");
            int choice = readInt("Choose: ");
            try {
                switch (choice) {
                    case 1:
                        students.addStudent(read("Roll number: "), read("Student name: "),
                            read("Department (CSE/ECE/IT/MECH etc.): "), readInt("Current semester: "));
                        break;
                    case 2:
                        academic.addCourse(read("Course code: "), read("Course title: "),
                            readDouble("Credits: "), read("Course type (THEORY/PRACTICAL/EMBEDDED): "));
                        break;
                    case 3:
                        academic.enroll(read("Roll number: "), read("Course code: "), readInt("Semester: "));
                        break;
                    case 4:
                        String ar = read("Roll number: "), ac = read("Course code: ");
                        int asem = readInt("Semester: "), held = readInt("Classes held: "), attended = readInt("Classes
attended: ");
                        boolean cond = read("Medical condonation approved? (y/n): ").equalsIgnoreCase("y");
                        academic.updateAttendance(ar, ac, asem, held, attended, cond);
                        break;
                    case 5:
                        String roll = read("Roll number: "), code = read("Course code: ");
                        int sem = readInt("Semester: ");
                        System.out.println("Enter a mark from 0-100, or press Enter to leave it unchanged.");
```

```
                Double cia1 = readOptionalMark("CIA Test 1: ");
                Double cia2 = readOptionalMark("CIA Test 2: ");
                Double assignment = readOptionalMark("Assignment/Quiz/Lab evaluation: ");
                String absentAnswer = read("Was the student absent for ESE? (y/n): ");
                boolean absent = absentAnswer.equalsIgnoreCase("y");
                Double ese = absent ? null : readOptionalMark("ESE mark: ");
                academic.updateMarks(roll, code, sem, cia1, cia2, assignment, ese, absent);
                break;
            case 6:
                academic.printMarkSheet(read("Roll number: "), readInt("Semester: "));
                break;
            case 7:
                academic.attendanceDefaulters();
                break;
            case 0:
                System.out.println("Exiting."); return;
            default:
                System.out.println("Invalid option.");
          }
       } catch (Exception e) {
          System.out.println("Operation failed: " + e.getMessage());
       }
    }
}

private static String read(String prompt) {
    System.out.print(prompt); return sc.nextLine().trim();
}
private static int readInt(String prompt) {
    while (true) try { return Integer.parseInt(read(prompt)); }
    catch (NumberFormatException e) { System.out.println("Enter a valid integer."); }
}
```

```
    private static double readDouble(String prompt) {
        while (true) try { return Double.parseDouble(read(prompt)); }
        catch (NumberFormatException e) { System.out.println("Enter a valid number."); }
    }
    private static Double readOptionalMark(String prompt) {
        while (true) {
            String s = read(prompt);
            if (s.isEmpty()) return null;
            try {
                double d = Double.parseDouble(s);
                if (d >= 0 && d <= 100) return d;
            } catch (NumberFormatException ignored) {}
            System.out.println("Enter a number from 0 to 100, or press Enter.");
        }
    }
```

## DBCONNECTION.JAVA:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/kec_academic";
    private static final String USER = "root";
    private static final String PASSWORD = "Subadharsh2008"; // Change this

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
```

**OUTPUT SCREENSHOTS:**

MAIN MENU:

```
===== KEC R2024 ACADEMIC SYSTEM =====
1. Add student
2. Add course
3. Enroll student in course
4. Update attendance
5. Phase 1: Enter CIA 1 and CIA 2
6. Phase 2: Enter assignment/quiz/lab mark
7. Phase 3: Enter ESE mark / mark AB
8. Generate semester mark sheet
9. Attendance defaulter report
10. Department result summary
0. Exit
```

## STUDENT & COURSE MANAGEMENT:

### 1. ADDING STUDENT:

```
Choose option: 1
Roll number: 25ITR125
Student name: Sangavi S
Department: IT
Current semester (1-8): 2
Student added successfully.
```

### 2. ADDING COURSE:

```
Choose option: 2
Enter course code: 24ITR31
Enter course name: Database Management Systems
Enter department: IT
Enter semester: 2
Enter credits: 4
Course added successfully.
```

### 3. STUDENT ENROLLMENT IN COURSE:

```
Choose option: 3
Enter roll number: 25ITR125
Enter course code: 24ITR31
Student enrolled successfully.
```

### 4. ATTENDANCE UPDATING FOR STUDENT:

```
Choose option: 4
Enter roll number: 25ITR125
Enter course code: 24ITR31
Enter classes held: 40
Enter classes attended: 36
Attendance updated successfully.
Attendance Percentage: 90.00%
```

### 5. MARKS UPDATING FOR STUDENT:

```
Choose option: 5
Enter roll number: 25ITR125
Enter course code: 24ITR31
Enter CIA 1 mark: 45
Enter CIA 2 mark: 48
CIA marks updated successfully.
```

## 6. MARKSHEET GENERATION:

```
Choose option: 8
Enter roll number: 25ITR125

===== SEMESTER MARK SHEET =====
Roll Number: 25ITR125
Student Name: Sangavi S
Department: IT
Semester: 2

Course Code: 24ITR31
Course Name: Database Management Systems
CIA 1: 45
CIA 2: 48
ESE Mark: 85
Result: PASS

Semester mark sheet generated successfully.
```

## 7. ATTENDANCE REPORT:

```
Choose option: 9

===== ATTENDANCE DEFAULTERS =====
Roll No: 25ITR125
Student Name: Sangavi S
Course Code: 24ITR31
Attendance: 60.00%
Grade: SA

Attendance defaulter report generated.
```

# DATABASE:

## STUDENTS DATABASE:

```
mysql> SELECT *FROM students;
+----------+--------------+------------+------------------+-------+-------+
| roll_no  | student_name | department | current_semester | email | phone |
+----------+--------------+------------+------------------+-------+-------+
| 25CSR120 | Deepeshwar   | CSE        |                3 | NULL  | NULL  |
| 25CSR173 | Mithun       | CSE        |                3 | NULL  | NULL  |
| 25ECR034 | Janani       | ECE        |                3 | NULL  | NULL  |
| 25ECR036 | Jaidharshini | ECE        |                3 | NULL  | NULL  |
| 25ECR045 | Lisha        | ECE        |                3 | NULL  | NULL  |
| 25ITR121 | Sahana       | IT         |                3 | NULL  | NULL  |
| 25ITR143 | Subadharsini | IT         |                3 | NULL  | NULL  |
| 25ITR144 | Subasree     | IT         |                3 | NULL  | NULL  |
| 25ITR153 | Swetha       | IT         |                3 | NULL  | NULL  |
| 25ITR164 | Trishna      | IT         |                3 | NULL  | NULL  |
| 25ITR172 | Vikasini     | IT         |                3 | NULL  | NULL  |
| 25MER001 | Akila        | MECH       |                3 | NULL  | NULL  |
| 25MER163 | Thulasi      | MECH       |                3 | NULL  | NULL  |
+----------+--------------+------------+------------------+-------+-------+
13 rows in set (0.00 sec)
```

## COURSES DATABASE:

```
mysql> SELECT *FROM courses;
+-------------+---------------------------------------+---------+-------------+------------+
| course_code | title                                 | credits | course_type | department |
+-------------+---------------------------------------+---------+-------------+------------+
| 24ADC31     | Artificial Intelligence & Knowledge   |     4.0 | EMBEDDED    | IT         |
| 24CEC31     | Surveying and Geomatics               |     4.0 | EMBEDDED    | IT         |
| 24CEL61     | Environmental Engineering Laboratory  |     1.0 | PRACTICAL   | IT         |
| 24CEP71     | Project Work Phase 1                  |     3.0 | PRACTICAL   | IT         |
| 24ITC33     | ITC                                   |     3.0 | THEORY      | IT         |
| 24ITT32     | Programming in Java                   |     3.0 | EMBEDDED    | IT         |
+-------------+---------------------------------------+---------+-------------+------------+
6 rows in set (0.00 sec)
```

## ENROLLMENTS DATABASE:

```
mysql> SELECT *FROM enrollments;
+---------------+----------+-------------+----------+--------------+-----------------+---------------------+
| enrollment_id | roll_no  | course_code | semester | classes_held | classes_attended | medical_condonation |
+---------------+----------+-------------+----------+--------------+-----------------+---------------------+
|             2 | 25ITR143 | 24ITT32     |        3 |           50 |              48 |                   1 |
|             3 | 25ITR153 | 24ITT32     |        3 |           50 |              25 |                   0 |
|             4 | 25ITR172 | 24ITT32     |        3 |           50 |              42 |                   0 |
|             5 | 25MER163 | 24CEP71     |        3 |           45 |              41 |                   0 |
|             6 | 25ITR121 | 24ITC33     |        3 |           40 |              38 |                   1 |
+---------------+----------+-------------+----------+--------------+-----------------+---------------------+
5 rows in set (0.00 sec)
```

## RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

## REFERENCES:


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**8. GeeksforGeeks, Establishing JDBC Connection in Java.** https://www.geeksforgeeks.org/establishing-jdbc-connection-in-java/ .

**9. YouTube, Java JDBC with MySQL Tutorial Videos.** https://www.youtube.com/results?search_query=Java+JDBC+MySQL+tutorial .

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## GITHUB:

| | |
|---|---|
| **LINK:** | https://github.com/sangavi2007-sd/JAVA-ASSIGNMENT |